Full End-to-End Data Analysis Project

Project Description

The objective of this project is to perform a complete end-to-end data analysis workflow using a real-world dataset from a domain of the student's choice, such as healthcare, finance, retail, HR, education, or supply chain.

Students are required to identify a business problem, analyze the data, extract meaningful insights, and provide actionable recommendations to support data-driven decision-making.

The selected dataset must contain **at least 5 related tables** that can be integrated and analyzed to answer relevant business questions.

Project Phases & Tools

1. Business Understanding & Data Collection

- Select a business domain and a suitable dataset.
- Define the business problem, objectives, and key analytical questions.
- Understand the dataset structure and relationships between tables.

2. Data Cleaning & EDA — Python

- Inspect data quality and handle missing values, duplicates, inconsistent data types, and outliers.
- Perform exploratory data analysis (EDA) to identify patterns, trends, and relationships.
- Create visualizations to support the analysis.
- Export the cleaned tables for database integration.

3. Database Design & SQL Analysis — MySQL or SQL Server

- Import the cleaned tables into the database.
- Design an ERD and relational schema as if building the database from scratch.
- Define primary keys, foreign keys, relationships, and appropriate data types.

- Write SQL queries using joins, aggregations, CTEs, subqueries, and window functions to answer business questions.

4. Data Modeling & Dashboard Development — Excel

- Build an appropriate data model and define KPIs.
- Develop two interactive dashboards:
  - **Dashboard 1: Overview** — High-level performance and key metrics.
  - **Dashboard 2: Deep Analysis** — Detailed analysis of a specific business area.

5. Data Modeling & Dashboard Development — Power BI

- Build an analytical data model and create relevant DAX measures.
- Develop two interactive dashboards:
  - **Dashboard 3: Deep Analysis** — Detailed exploration of another important aspect of the data.
  - **Dashboard 4: Deep Analysis** — Detailed exploration of another important aspect of the data.

6. Insights & Recommendations

- Interpret the analytical results and dashboard findings.
- Identify significant patterns, business challenges, and opportunities.
- Provide evidence-based, actionable recommendations.

7. Final Presentation

- Present the business problem, dataset, methodology, database design, key findings, dashboards, and recommendations.

Final Deliverables

1. Original dataset files.
2. Python notebook (`.ipynb`) containing data cleaning and EDA.
3. SQL file (`.sql`) containing database design and analytical queries.
4. ERD and relational schema.
5. Excel workbook (`.xlsx`) containing two interactive dashboards.
6. Power BI report (`.pbix`) containing two interactive dashboards.
7. PDF report summarizing insights and recommendations.
8. PowerPoint presentation (`.pptx`) covering the complete project.

General Requirements

- The dataset must contain at least five related tables.
- The chosen domain and business problem must be clearly defined.
- Each phase must contribute to answering the business questions.
- All dashboards must be interactive, relevant, and business-oriented.
- Insights and recommendations must be based on actual analytical findings.
- Results and KPI definitions should be consistent across Python, SQL, Excel, and Power BI.